

Worksheet 8B: Probability Distributions — Solutions

Series 8: Probability Deepened
AIML Foundations Mathematics
Dublin and Dún Laoghaire ETB
Instructor: Josh Aaron

Instructor Copy — Not for Distribution

Note to instructor: Key answers are given. Accept mathematically equivalent forms.

Part A

- (a) $\binom{5}{3}(0.6)^3(0.4)^2 = 10(0.216)(0.16) = 0.3456$ (b) $P(0) = 0.01024$, $P(1) = 0.07680$, $P(2) = 0.23040$, $P(3) = 0.34560$, $P(4) = 0.25920$, $P(5) = 0.07776$; sum = 1✓. (c) Mean = 3, Variance = 1.2.
- (a) $\binom{10}{8}(0.8)^8(0.2)^2 = 45(0.1678)(0.04) \approx 0.302$ (b) $P(9) + P(10) = \binom{10}{9}(0.8)^9(0.2) + \binom{10}{10}(0.8)^{10} \approx 0.268 + 0.107 = 0.376$ (c) Mean = 8, SD = $\sqrt{1.6} \approx 1.265$.
- (a) $(p + (1-p))^n = 1^n = 1$; expand by binomial theorem to get $\sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k} = 1$. ✓
- (a) 10 (b) 1 (c) 120 (d) 70
- (a) Binomial(20, 0.1) (b) $(0.9)^{20} \approx 0.1216$ (c) $P(0) + P(1) + P(2) \approx 0.1216 + 0.2702 + 0.2852 = 0.677$ (d) $np = 2$
- (a) 10 (or 10 — symmetric, both are modes) (b) 10 (c) Approaches a bell shape (Normal) by CLT.
- Mean = 30, Variance = 21, SD ≈ 4.58 .
- (a) $500 \times 0.02 = 10$ (b) $\binom{500}{15}(0.02)^{15}(0.98)^{485}$ (c) 15 is 50% above expected; a Poisson approximation ($\lambda = 10$) gives $P(X \geq 15)$ which is in the tail — suggests possible difference.
- $P(X = 3) = (0.75)^2(0.25) \approx 0.141$; $P(X = 4) = (0.75)^3(0.25) \approx 0.105$; $P(X = 5) \approx 0.079$.
- Mean = $1/0.1 = 10$; Variance = $0.9/0.01 = 90$.
- Poisson($\lambda = 2$).
- (a) True (b) True (c) True when $p > 0.5$: e.g. $n = 10, p = 0.9$: mean 9, var $0.9 < 9$... actually variance = $np(1-p) = 0.9 < \text{mean} = 9$. Variance < mean always for Binomial. (d) True.

Part B

- (a) $\frac{4^4 e^{-4}}{4!} = \frac{256}{24} e^{-4} \approx 0.1954$ (b) $e^{-4} \approx 0.0183$ (c) $1 - \sum_{k=0}^6 P(k) \approx 1 - 0.889 = 0.111$ (d) $\lambda = 4$

14. (a) $e^{-2} \approx 0.135$ (b) $\frac{4e^{-2}}{2} \approx 0.271$ (c) $e^{-2}(1 + 2 + 2) = 5e^{-2} \approx 0.677$ (d) $1 - P(0) - P(1) - P(2) - P(3) = 1 - 5e^{-2} - \frac{8e^{-2}}{6} \approx 1 - 0.677 - 0.180 = 0.143...$ check: $P(X > 3) = 1 - P(0) - P(1) - P(2) - P(3) \approx 0.143$.
15. (a) $\frac{10^8 e^{-10}}{40320} \approx \frac{10^8 \times 0.0000454}{40320} \approx 0.1126$ (b) $\lambda = 5$ patients (c) $10/60 \approx 0.167$ per minute.
16. (a) $e^{-0.5} \approx 0.607$; $0.5e^{-0.5} \approx 0.303$; $\frac{0.25e^{-0.5}}{2} \approx 0.076$ (b) $1 - e^{-0.5} \approx 0.393$
17. $\lambda = 2$; $P(X = 0) = e^{-2} \approx 0.135$; $P(X = 2) = 2e^{-2} \approx 0.271$. Binomial: $P(0) = (0.99)^{200} \approx 0.134$; $P(2) = \binom{200}{2}(0.01)^2(0.99)^{198} \approx 0.273$. Good approximation.
18. (a) $e^{-3} \approx 0.0498$ (b) $\lambda = 1.5$ (c) $e^{-1.5} \approx 0.223$
19. $P(X \geq 1) = 1 - e^{-0.3} \approx 1 - 0.741 = 0.259$
20. Binomial: $P(0) = (0.94)^{50} \approx 0.0453$; $P(1) = 50(0.06)(0.94)^{49} \approx 0.1445$; $P(2) = \binom{50}{2}(0.06)^2(0.94)^{48} \approx 0.2250$. Poisson(3): $P(0) = e^{-3} \approx 0.0498$; $P(1) = 3e^{-3} \approx 0.149$; $P(2) = \frac{9e^{-3}}{2} \approx 0.224$. Very close.
21. $e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} \cdot e^{\lambda} = 1$. ✓
22. (a) $\lambda = 3$ calls in 15 min (b) $e^{-1} \approx 0.368$ (c) Exponential distribution.

Part C

23. (a) $f(x) = 0.1$ for $0 \leq x \leq 10$, horizontal line. (b) $(7 - 3)/10 = 0.4$ (c) $2/10 = 0.2$ (d) Mean = 5, Variance = $100/12 \approx 8.33$.
24. (a) $3/10 = 0.3$ (b) 5 minutes.
25. (a) $1/\lambda = 10$ min (b) $e^{-0.1(15)} = e^{-1.5} \approx 0.223$ (c) $1 - e^{-0.1(5)} = 1 - e^{-0.5} \approx 0.393$
26. $P(X > s + t \mid X > s) = \frac{P(X > s+t)}{P(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t)$. ✓
27. (a) $\text{Exp}(\lambda = 4)$ (b) $e^{-4(1)} = e^{-4} \approx 0.0183$ (c) $1/4 = 0.25$ min.
28. $\int_0^{\infty} \lambda e^{-\lambda x} dx = [-e^{-\lambda x}]_0^{\infty} = 0 - (-1) = 1$. ✓
29. (a) Poisson (b) Exponential (c) Uniform (d) Poisson
30. (a) 100 hours (b) $e^{-0.01(100)} = e^{-1} \approx 0.368$ (c) $1 - e^{-0.01(50)} = 1 - e^{-0.5} \approx 0.393$

Part D

31. (a) $[170, 186]$ (b) $[162, 194]$ (c) $194 = \mu + 2\sigma$: 2.5% above (d) $z = (162 - 178)/8 = -2$
32. (a) $65 + 1.5(10) = 80$ (b) 68% (55 to 75 is $\pm 1\sigma$) (c) $\mu + 2\sigma = 85$
33. (a) $z = (9.8 - 10)/0.1 = -2$ and $z = +2$ (b) 95% (c) Reduce σ (tighter manufacturing).
34. (a) $P(X < 55) = P(Z < 1) \approx 0.841$ (b) $P(Z < 2) - P(Z < -1) = 0.977 - 0.159 = 0.818$ (c) $50 + 2(5) = 60$

35. The log-likelihood under Gaussian noise is $\sum[-\frac{\epsilon_i^2}{2\sigma^2} - \frac{1}{2} \ln(2\pi\sigma^2)]$; maximising over predictions (fixing σ^2) is equivalent to minimising $\sum \epsilon_i^2$, i.e. MSE.
36. Median: $P(\ln X \leq m) = 0.5 \Rightarrow m = e^\mu = e^2 \approx 7.39$. Right-skewed because the distribution has a long right tail: large values of X correspond to $\ln X$ near the upper tail of the normal.
37. (a) True (b) False ($P(X = c) = 0$ for all c) (c) False (d) True
38. $X_1 \sim N(2, 9)$, $X_2 \sim N(3, 16)$: $X_1 + X_2 \sim N(5, 25)$. SD of sum = 5 $\neq \sqrt{9} + \sqrt{16} = 7$. Variances add; standard deviations do not.

Part E

39. (a) $S \sim N(350, 100 \times 35/12) \approx N(350, 291.7)$ (b) $P(S > 370) = P(Z > 20/\sqrt{291.7}) \approx P(Z > 1.17) \approx 0.121$ (c) SD of $\bar{X} = \sqrt{35/1200} \approx 0.171$; $P(|Z| < 1.17) \approx 0.758$.
40. (a) $N(40, 24)$ ($\mu = 100(0.4) = 40$, $\sigma^2 = 100(0.4)(0.6) = 24$) (b) $P(X \leq 45) = P(Z \leq 1.02) \approx 0.846$ (c) Most accurate when $np \geq 5$ and $n(1-p) \geq 5$.
41. (a) $N(3000, 4500)$ (SD ≈ 67.1) (b) $P(S > 3100) = P(Z > 100/67.1) \approx P(Z > 1.49) \approx 0.068$.
42. By the CLT, the average model's error variance is σ^2/n : each additional tree reduces variance by a factor of $1/n$. With $n = 100$ trees, variance is $100\times$ smaller — dramatically improving stability and accuracy.
43. Correlated stock prices: $\bar{X}$ of returns doesn't satisfy CLT assumptions since returns in the same market are correlated. Also: autocorrelated time series (e.g. daily temperatures).
44. (a) $X \sim \text{Binomial}(n, p)$ is the sum of n independent Bernoulli(p) variables. (b) A Poisson process also has the memoryless property: past arrivals give no information about future arrivals (like a Markov chain with state = "current count"). (c) Normal($0, \sigma^2$) noise $\Rightarrow$ minimising $\sum \epsilon_i^2$ (MSE) is maximum likelihood. (d) By the CLT, $\bar{x} \sim N(\mu, \sigma^2/n)$: the sample mean is approximately normally distributed, so confidence intervals and t -tests are justified.